AADITYA SINGH RAWAT

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EDUCATION

VIT Bhopal University (September 2023 – Present)

B. Tech CSE | CGPA: 8.82

SKILLS

- **Languages:** Java, Python, JavaScript, SQL, HTML, CSS
- **Frameworks & Libraries:** Node.js, Express.js, React.js, Tailwind CSS, Mongoose.
- **Databases & Tools:** MongoDB, AWS, Git

PROJECTS

Asynchronous Notification Engine

- Architected a high-throughput, asynchronous backend system using Node.js and TypeScript to decouple HTTP request ingestion from background processing and prevent server blocking.
- Implemented Redis-backed message queues via BullMQ featuring strict idempotency controls and automated Time-To-Live (TTL) expiration to eliminate duplicate dispatches.
- Enforced edge-level payload validation using Zod and integrated Nodemailer with SMTP providers for reliable transactional notification delivery.
- **Tech Stack:** Node.js, TypeScript, Express.js, BullMQ, Redis, Zod, Nodemailer, Docker.

QueryOptima – Full-Stack Database Performance & API Analytics Dashboard

- Engineered a centralized developer analytics platform using the MERN stack to intercept, track, and visualize API latency and database query performance metrics.
- Developed a custom Express.js tracking middleware that benchmarks individual API endpoint execution times down to the millisecond, logging performance metrics asynchronously to prevent server overhead.
- Optimized dashboard data-retrieval speeds by 42% by leveraging advanced MongoDB Aggregation Pipelines (bucket, match, group) to bucket API latencies into historical time-series data**.**
- **Tech Stack:** MongoDB, Express.js, React.js, Node.js, Recharts, Tailwind CSS, Mongoose**.**

CERTIFICATION

IIoT Markets and Security certification by Coursera (University of Colorado Boulder)

- basics of Industrial IoT real-time OS security, network and wireless comm. protocols

Blockchains and its Applications by NPTEL

- in depth knowledge and hands on practice of cryptographic methods and security methods

AWS Technical Essentials – Amazon Web Services (AWS)

- Fundamentals of AWS cloud services and infrastructure

EXTRA-CURRICULARS AND ACHIEVEMENTS

ACHIEVEMENTS

- Engaged with community discussions during the **100 Days of Code** project, sharing insights after completing each milestone; cultivated collaborative learning habits that enhanced overall understanding among peers involved in similar initiatives.

EXTRACURRICULAR

- **Lead of Technical Team, Pahadi Club – VIT Bhopal** Leading the technical team of the Pahadi club, promoting the culture and handling technical provisions and issues in all the events done in the college.
- **National Service Scheme - VIT Bhopal University –** Participated in the plantation camp where the team successfully planted 10,000 trees, contributing to environmental sustainability and community awareness.

ADDITIONAL INFORMATION

- **Hobbies:** Solving problems, Gaming, Reading novels.
- **Languages:** English, Hindi.

Leetcode